

NETFRAME

Operations Documentation Disaster Recovery Guide

Recovery objectives, service tiers, and ordered procedures
for partial and total loss

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A redacted public sample from the NetFRAME operations library.
Demonstrates operational-documentation discipline; not an operational document.

github.com/machismo0311/netframe-ops-docs-public

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This is the Markdown edition of the guide, provided so it renders natively on GitHub. The **PDF edition** is the typeset original; both carry the same content. This is a **redacted public sample**: addresses and hostnames are generalized to roles (GPU-A, STORAGE, NODE-N), and container/VM IDs, service ports, exact versions, and internal repository names are omitted. It is not an operational document. The full internal edition is kept private.

Purpose. Provide recovery scenarios, objectives, and ordered procedures for partial and total loss.

Scope. Node loss, storage loss, firewall loss, and total-site loss. The full bare-metal build sequence is in NF-RC-010; restore mechanics are in NF-RC-001.

Assumptions. Backups are current and verified (dead-man switch alert active). PBS lives on STORAGE.

Prerequisites. Physical/console access; the out-of-band secrets bootstrap (see below and NF-OP-030).

1. Recovery objectives (RTO / RPO)

Service tier	RTO	RPO	Basis
Edge (OPNsense)	Under 1 h	Last config backup	Verified serial console + encrypted config repo
DNS (Pi-hole x2)	Under 1 h	15 min	Secondary + sync mirror
Storage / PBS (STORAGE)	4 to 24 h	24 h	Single node; rebuild + restore
Cluster guests (VMs/CTs)	2 to 8 h	24 h	PBS restore, manual (HA not arming guests)
Kubernetes (RKE2)	4 to 24 h	Varies	Rebuild CPs; PV data depends on STORAGE
Secrets (the secret store)	Bootstrap-gated	n/a	See circular-dependency break below

Note. These objectives are targets to validate by drill, not guarantees. The restore path is not yet rehearsed (see NF-RC-001); the first drill should confirm or correct these numbers.

2. Scenario: single node loss

Quorum tolerates up to three node losses (7 votes, needs 4). Recover: restore the node's guests from PBS to a healthy node (NF-RC-001), or rebuild the node (NF-RC-010) and restore. Do not force quorum. If the lost node hosts a RKE2 control-plane VM, restore it only after confirming the remaining etcd members are healthy (2 of 3).

3. Scenario: STORAGE (storage) loss

This is the highest-impact partial failure: all PBS backups, Kubernetes NFS storage, the registry PV, and NFS exports are on STORAGE. Recovery: rebuild STORAGE (NF-RC-010), import the ZFS pools (or rebuild from disks), and restore PBS. Until STORAGE is back, Kubernetes dynamic provisioning and cluster backups are unavailable.

Caution. Mitigate before it happens: add an offsite copy of the critical backup subset (an offsite object store or a second PBS) so STORAGE is not the only place backups exist. This is a documented DR gap.

4. Scenario: OPNsense (firewall) loss

Loss removes internet, inter-VLAN routing, DHCP, and DNS recursion. Recover: install OPNsense from ISO on a cluster node, import the latest encrypted config from the config backup repo (the age key is required, see below), and verify all VLAN/firewall rules. The verified serial console procedure and cold-restore runbook target minutes, not hours.

5. Scenario: total-site loss (bare-metal rebuild)

Execute NF-RC-010 (Build Guide) in order. The critical first step is the secrets bootstrap.

5.1 The secrets circular dependency (must solve first)

All secrets live in the secret store, which runs as a container inside the cluster being rebuilt. During a total loss you cannot unlock the secret store because the cluster does not exist yet.

Caution. Maintain an out-of-band, encrypted bootstrap bundle stored offline (not in the cluster): the secret store export (or master credential), the age key for the OPNsense config repo, and the restic repository password. Without this bundle a total-site rebuild cannot complete. See NF-OP-030.

5.2 Rebuild order (summary)

Bootstrap secrets -> Proxmox on each node -> cluster/quorum -> OPNsense (network) -> Pi-hole DNS -> STORAGE storage + PBS -> core LXC's (the secret store first to close the loop) -> RKE2 -> GPU nodes. Each phase has a validation gate in NF-RC-010.

6. Troubleshooting

Symptom	Action
Cannot unlock the secret store during rebuild	Use the out-of-band bootstrap bundle; do not block the rebuild waiting for it.

Symptom	Action
PBS restore fails (no backups reachable)	Confirm STORAGE pools imported; check the STORAGE PBS datastore; fall back to offsite copy if present.
DNS down cluster-wide after edge restore	Confirm the edge resolver and both Pi-hole upstreams; see NF-OP-004.
Quorum will not form	Bring up 4 of 7 nodes; check corosync and time sync; never force before that.

See also: NF-RC-001 Backup/Restore; NF-RC-010 Build Guide; NF-OP-030 Identity and Secrets; NF-AR-030 Storage; NF-GV-001 Risk Register.